Swing Electrical Power Generator

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Declaration

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Abstract

This project presents the design, simulation and the implementation of a swing electrical power generator which converts kinetic energy produced by a back-and-forth motion of a swing to usable electrical energy. The aim of this project is to find different methods of small-scale renewable energy generation using kinetic energy in an everyday occurrence. The system makes use of a rotational conversion component and an electrical generator to convert kinetic oscillatory motion into an electrical output. Simulation tools will be used to model the systems performance, followed by a practical implementation and testing to evaluate the efficiency of the system. This project highlights the potential of harvesting renewable energy into contributing to low-cost and low-power applications.

Table of Contents

List of Figures

List of Illustrations

List of Tables

1. Introduction

1.1 Background to the study

Energy is the ability to do work. It can be converted from one form to another such as heat, light, motion, chemical and electrical. In recent years, energy consumption has increased vastly due to population growth, economic development and industrialization, technological advancements, climate and environmental factors [2]. In South Africa, the leading contributor towards electricity generation is coal through the conversion of the chemical energy stored in coal. In the year 2024, coal was responsible for approximately 82% of the country's electricity generation [3]. Coal has the highest percentage of energy generated. However, the constant reliance on coal brings challenges such as long-term sustainability, poses health risks, pollution and rising costs due to market demand.

Despite the high energy production, South Africa still faces an energy crisis driven by aging coal infrastructure, mismanagement at Eskom, who is responsible for over 80% of electricity generation, and insufficient generation capacity [1].

This is where renewable energy stands out. Renewable energy is energy that never runs out and can be used for a wide variety of applications, such as solar, wind and hydro.

To help meet this demand, engineers found a solution by using renewable energy. Renewable energy is energy derived from natural sources that are naturally replenishing, such as wind, solar and kinetic energy. Unlike solar and wind conditions which depend on the weather to generate electricity, human-powered energy generation will occur while people use the equipment. In this project, it is proposed to generate electrical energy from the kinetic energy produced by the oscillatory motion of a playground swing that will be low cost, low-resource, low maintenance, safe for kids and environmentally friendly. therefore, harvesting human-powered energy to produce electricity. This study investigates the design, simulation and implementation of the swing-based electrical power generator.

1.2 Objectives of this study

1.2.1 Problem Statement

The growing demand for electricity as well as the environmental and economic setbacks of fossil-fuel generation, has created a need for small-scale creative and innovative renewable energy solutions.

While large-scale solar and wind renewable energy is widely pursued, there still remains a significant gap in the development of low-cost energy harvesting systems that will be suitable for communities and educational applications. Kinetic energy produced during human activity in public spaces such as parks and playgrounds is part of a wide energy source that has been untouched. The pendulum-like motion that is generated by a person swinging contains mechanical energy which can be harvested and converted into electrical energy. This can serve as a practical and sustainable power source for low-power applications such as park lighting and phone charging station. Although the challenge lies in effectively converting the slow, bidirectional and irregular motion of a swing into smooth and usable electricity.

This study investigates the following key questions:

- How can the oscillatory kinetic energy of a swing be effectively converted to electrical energy?
- What mechanical design and configuration best optimises energy transfer from swing motion to the generator?
- Can a simulation model accurately predict the electrical energy generated from the system based on swing dynamics and operating conditions?
- Is the prototype capable of producing a sufficient electrical output for low-power application?

1.2.2 Purpose of the study

i. ***Energy access and community relevance***

South Africa's national electricity access rate is approximately 84.4%, where rural areas have a much lower access rate of 75.3% [14]. This means that majority of the population, especially rural and underserved communities remain without reliable access to electricity. This is where this project of a proposed swing-based generator comes in, to serve as a small-scale energy harvesting system which offers as a practical and low-cost solution for supplying electricity to public spaces and off-grid communities that cannot be easily accessed by the conventional grid infrastructure.

ii. ***Environmental significance***

The continue dominance of coal in South Africa's energy contributes significantly to greenhouse gases and air pollution, which brings about serious risks to public health and the environment. Energy harvesting devices can make use of renewable energy sources that are self-replenishing and abundant such as solar, wind, thermal and kinetic energy, which reduce the need for power transmission that cause energy losses and environmental impacts. Hy harnessing human kinetic energy, the swing generator being investigated in this study produces no harmful emissions and requires no fuel, making it an environmentally safe solution for electricity generation [15].

iii. ***Educational and social impact***

Beyond the technical contribution, this study can be a major educational tool. Swings are enjoyed by people of all ages, making them an effective tool for demonstrating principles of energy conversion in a physical and engaging way. When the swing powers a light or a device in a park, it makes the concept of renewable energy visible for communities to observe and understand.

1.3 Scope and Limitations

<u>Scope</u>

This study focuses on the design, simulation and implementation of a small-scale swing-based electrical power generator that converts kinetic energy from the oscillatory motion of a playground swing into usable electrical energy. The scope of this study includes:

- The mechanical design of the energy conversion system.
- The electrical design of the output circuit.
- Development of the simulation model using Matlab/Simulink to predict the electrical output based on swing dynamics and operating conditions.
- Building a prototype of the swing generator.
- Evaluating the prototypes suitability for low-power applications such as powering a light.

<u>Limitations</u>

- Time constraints: The duration of this study is limited to one academic year which leaves less time for continuous testing and improvements.
- Budget constraints: The prototype is limited to low-cost components, which may not represent the best components that could be used in a real-life adaptation.
- Prototype scale: This study is for a small-scale lab model. Results may vary than a full-scale adaptation in a real playground.

2. Literature Review

2.1 Introduction to Renewable Energy Sources

Renewable energy is defined as the usable energy harvested from resources that are naturally replenishing such as solar, wind, hydro, geothermal heat, biomass [4]. Unlike fossil fuels such as coal, oil and natural gas, which accumulated after millions of years to form and are finite in supply, renewable energy sources are continuously replenished by nature and do not deplete over time [5]. An easily accessible source of renewable energy is a swing. Swings move in a pendulum-like motion, producing kinetic energy that can be transformed into electricity. Khan and Saeed et al. [6] indicates that it is possible to access this free energy source without using fossil fuels by placing energy harvesting systems on swings in parks and playgrounds.

Mullane et al. [7] study states that implementing swing-based kinetic energy systems demonstrates the usefulness of renewable energy in a hands-on manner. According to [8], powering lighting and small electronics through swings reduces the area's burden on the traditional grid. Swings are enjoyed by people of all ages, which makes them a great tool for teaching people about the importance of saving energy and protecting the environment. When a swing is used to power lights or small devices in a park, it can help communities become more aware of environmental issues and encourage communities to think of new and innovative ways to generate energy.

2.2 Swing Power Generations

Swing power generation refers to the process of converting kinetic energy from the swinging motion of a swing and converting it into electrical energy [9]. This concept has gained attention through its high potential for generating sustainable electricity from human activities, in playgrounds and public spaces to be specific. The idea is to capture the kinetic energy produced by the oscillatory motion and convert it into electrical power for low-power applications.

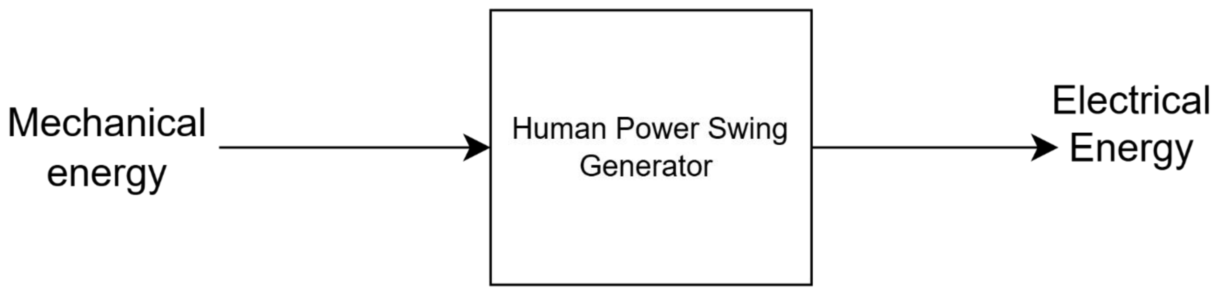


Figure 1: Simple block diagram showing conversion of energy

Riemer and Shapiro [10] explain the simple system behind power generation which involves attaching a generator or other energy harvesting devices to the joints of the swing and as the swing moves back and forth, the motion causes the generator to spin and produce electricity. Swing power systems can make use of different types of swings to capture energy.

- Simple playground swings with hanging seats can generate power [11].

- More advanced pendulum-style swing that uses counterweights and fuller arm motions [11].

Some designs use dual generators at the top and bottom pivots of the swing to harness energy in both directions of motion. The study of swing motion by Shastri and Bharath [12], investigates two different designs to drive a mechanical and electrical load:

- Harnessing bi-directional motion of the swing.
- Harnessing any one direction of the swing.

Although, some engineering challenges include optimising the generator design for varying swing speeds as well as building a durable system to withstand weather conditions and years of use. Mortley [13] describes how to build a swing set that generates electricity by using an old swing set by hanging a drive shaft below the top of the swing and attaching swings to it by roller clutches. As the swing begins motion, it would turn the shaft forward, and when swung backwards, the roller clutches will keep motion, and the momentum of the generator keeps forward motion. As the motions continues back and forth, the generator generates electricity.

2.3 Existing Research on Swinging Systems for Generating Energy

This section will discuss the studies that were done on swing systems.
Goku et al. [16] researched possible ways to reduce the high demand of power in India by developing a pendulum power generator, which will convert mechanical energy (kinetic) from a swinging pendulum weight to generate electricity. Ayneendra et al. [17] discussed using a swing to develop power in a system where mechanical energy is provided through a swing and is turned into electrical energy by a commutator and then stored within a battery. In order to transmit motion to the free wheel, which will rotate based on the swings angle of motion, the shaft will be combined to a sprocket. A chain will connect the sprocket and free wheel to convert angular motion into a whole rotation. The shaft that is attached to the free wheel will rotate the generator to produce the electricity.
Manjunatha et al. [18] developed a children's swing to play, that while in use it generates power. Anandhu, et.al. [19] had done a study which included a swing that uses sea waves to generate electricity as it is in use. The swing's simple structure causes the horizontal beam member to swing through an angle and continue to move back and forth.

3. Methodology

3.1 Research Approach

This study follows an experimental engineering approach, along with theoretical design, computer-based simulation and practical prototype construction to investigate the effect of harvesting kinetic energy from a swing.

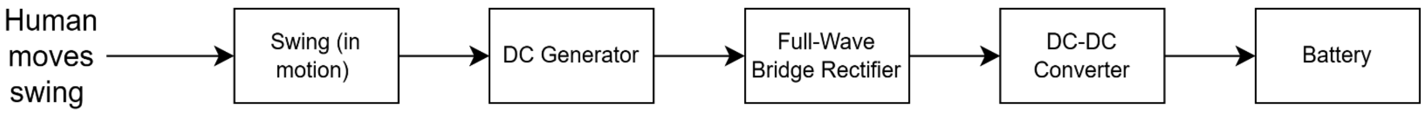


Figure 2: Block diagram showing the supposed system

Figure 3 shows the simplified block diagram of the proposed system for the swing power generator.

3.2 Specifications

After much research, it was determined that swings can be powered in various ways such as being human powered, driven by a motor and wind power.

Table 3-1: Comparison of Power supply:

Name:	Human Powered Swing:	Motor Driven swing:	Wind Powered swing:
Description:	A person sits on the swing and uses their body to make it move back and forth.	An electric motor is used to push the swing back and forth automatically.	Wind pushes the swing back and forth through a sail or paddle attached to the swing.
Advantage:	No fuel or electricity is needed. Anyone can use it and it teaches people about renewable energy in an interactive way.	The swing moves at a steady and controlled speed every time.	No person or motor is needed. It runs on its own as long as there is wind.
Disadvantages:	No power is produced when nobody is using the swing.	A power source is needed to run the motor, which means the system uses more electricity than it produces.	It only works when there is enough wind.
Why not chosen?	This method was chosen because it fits the criteria.	Using a motor to generate electricity would use more electricity than it produces, therefore defeating its purpose.	Wind is not always available.

Therefore, after considering different types of ways to cause the swing to move, it has been finalised to use a human-powered source for this project to generate the kinetic energy in the swing.

3.3 Main components of this model:

- Swing frame
- Generator
- Free wheel(large)
- Generator Sprocket (small)
- Roller chain
- Full wave bridge rectifier
- DC-DC converter
- Rechargeable Battery
- Pillow block bearings
- Roller clutch bearings

3.3.1 Swing Frame:

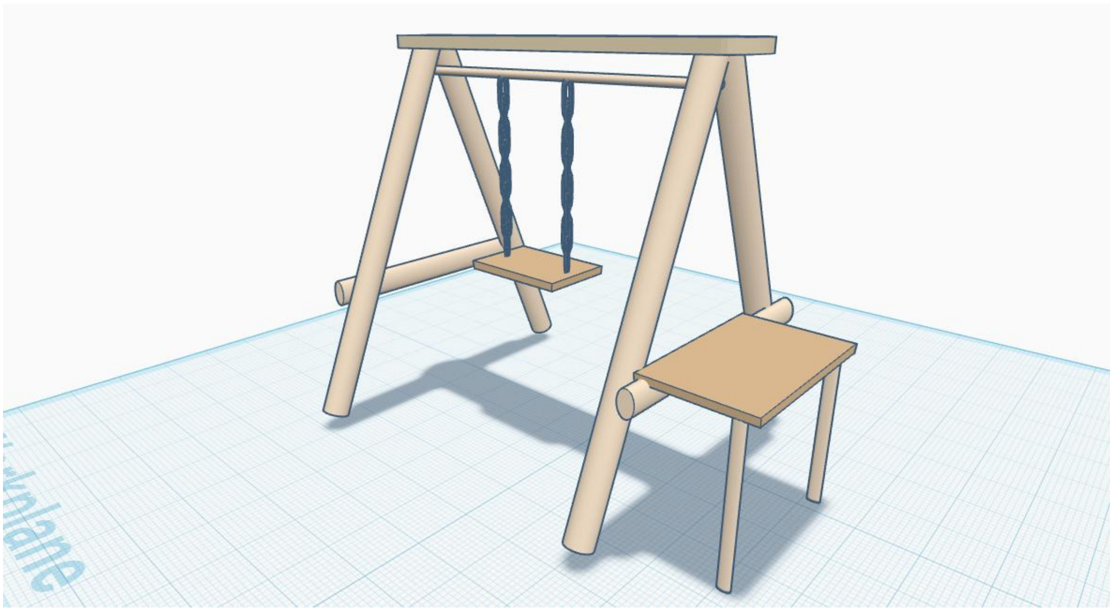


Figure 3: Simulation of swing frame

Table 3-2: Comparison of Swing Frame materials:

Name:	Metal frame:	Wooden Frame:	3D Printed Frame:
Description:	Made from aluminium that is welded or bolted together.	Made from wood that is joined together using screws or bolts.	Designed on a computer and printed using a 3D printer
Advantage:	Very strong and lasts a long time even under heavy use.	Simple to build using basic tools that is accessible.	Can be designed to exact measurements.
Cost:	Expensive.	Cheap.	Cheap.
Why not chosen:	Too heavy and expensive. Welding is not accessible.	This method was chosen.	Too weak for this prototype.

After comparing the three options of materials that can be used for the swing frame, the wooden frame has been finalised as the chosen method for this system prototype. It was chosen due to wood being cost-effective and construction made easier. Each leg beam will be 2100mm. The cross beam is 1700mm. The support beam at the top is 1500mm. The steel shaft is 1700mm.

3.3.2 Generator:

Table 3-3:Comparison of different options of Generators

Name:	Permanent Magnet DC Generator:	Stepper Motor Used Generator:	Brushless DC Generator:
Description:	Uses permanent magnets to produce electricity when the shaft is rotate.	Stepper motor that's repurposed to act as a generator by rotating its shaft to produce voltage.	Uses electronic commutation instead of brushes to produce electricity.
Advantage:	Simple to use.	Cheap and easy to find.	Very efficient and durable with no brushes to wear down.
Cost:	Low to moderate.	Very cheap.	Moderate to high.
Why not chosen:	This component was chosen.	Output voltage is irregular and difficult to control without additional processing circuits.	More expensive and requires additional electronic speed controllers.

After considering three different types of generator options, it was concluded that the Permanent Magnet DC Generator will be utilised in this swing system. This generator was chosen due to easy implementation, affordable and best fitting for this system. The rating is 12V 320 RPM DC Motor.

3.3.3 Free wheel(large):

Table 3-4: Comparison of different material Freewheels

Name:	Semicircle Plywood Freewheel:	Metal Freewheel:
Description:	A semicircle shaped freewheel cut from plywood.	A metal freewheel that's used to convert back and forth motion into continuous rotation.
Advantage:	Cheap and easy	Strong, durable and smooth operation under load.
Cost:	Very cheap.	Moderate.
Why not chosen:	Too weak and prone to warping under repeated stress.	This component was chosen.

After considering both freewheel options, it was decided that the metal freewheel will be used. The semicircle plywood freewheel was considered due to the low cost and ease of acquiring but was not chosen since it was too weak for this system. The metal freewheel is a much more reliable solution. The freewheel will be 80 Teeth 25H.

3.3.4 Generator Sprocket (Small):

After considering two methods of combining the large freewheel and the generator, it was decided to make use of a sprocket and chain system. Connecting the pulley directly to the generator shaft was not chosen because it requires higher rotational speed than what the swing and freewheel will produce, therefore there will be insufficient RPM provided to generate electricity. The sprocket system can be designed to produce the needed RPM by choosing different sizes of the sprocket to either step up or step down the speed. The sprocket will be 7 Teeth 25H.

3.3.5 Roller chain

Table 3-5: Comparison of different types of roller chains

Name:	Rope:	Rubber Band Pulley:	Roller Chain:
Description:	A rope looped around two pulleys to transfer rotational motion.	A rubber band around two pulleys to transfer motion.	A metal chain that engages with sprocket teeth to transfer rotational motion.

Advantage:	Cheap and easy to source.	Very cheap and simple to set up.	No slipping.
Cost:	Cheap.	Cheap.	Moderate.
Why not chosen:	Prone to slipping and stretching under load.	Too weak, unable to handle the required load.	This component was chose.

3.3.6 DC-DC Converters:

Table 3-6: Comparison of different types of DC-DC Converters:

Name:	Buck Converter:	Buck-boost Converter:	Boost Converter:
Description:	Steps voltage down to a stable lower level.	Can both step voltage up and down depending on the input voltage.	Steps voltage up to a higher level than the input.
Advantage:	Very efficient at converting higher voltages down.	Handles variable input voltage in both directions.	Useful when the generator output voltage is too low and needs to be increased.
Cost:	Cheap.	Moderate.	Cheap.
Why not chosen:	Cannot handle input voltage below the target output since swing speed varies so voltage drops below 12V.	More expensive and unnecessarily complex since the generator voltage never exceeds 12V.	This method is chosen.

After evaluating all three voltage regulation options, it was decided that the boost converter would be used in the swing power generator system. The buck converter was ruled out as the swing moves at varying speeds causing the generator output voltage to frequently drop, which a buck converter cannot step up. The buck-boost converter was not used as it is more expensive and complex for this system since the generator is driven by a slow-moving swing it will never produce a voltage exceeding 12V. This system will make use of an XL6009 boost converter module.

3.3.7 Full Wave Rectifier Selection

For this project a single package rectifier module was selected as the most suitable rectifier for the electrical output circuit. The rectifier module has four diodes assembled inside one component, making it easy to connect directly to the generator output and the battery charging circuit. Since the swing produces motion in both directions, the rectifier module converts both directions of current into a single direction of DC current suitable for charging the battery. The module is cheap, widely available and reduces the risk of loose connections that could occur during the vibration caused by the swing motion. For this system, W10M Bridge Rectifier will be used.

3.3.8 Battery:

Table 3-7:Comparison of different types of Rechargeable Batteries:

Name:	Lead Acid Battery	Lithium Ion Battery	Nickel Metal Hydride Battery
Description:	Heavy rechargeable battery commonly used in cars	Lightweight rechargeable battery commonly used in phones	Rechargeable battery commonly used in household devices
Advantage:	Cheap and can handle being overcharged.	Lightweight and stores a large amount of energy.	Safe, reliable and widely available.
Cost:	Very cheap.	Expensive.	Moderate.
Why not chosen:	Too heavy and bulky for a small-scale prototype.	Too expensive for a small-scale prototype and requires a special charging circuit to prevent overcharging and damage.	This component was chosen.

For this system, I will be using a 12V 10Ah Sealed Lead Acid battery.

3.3.9 Pillow block bearings:

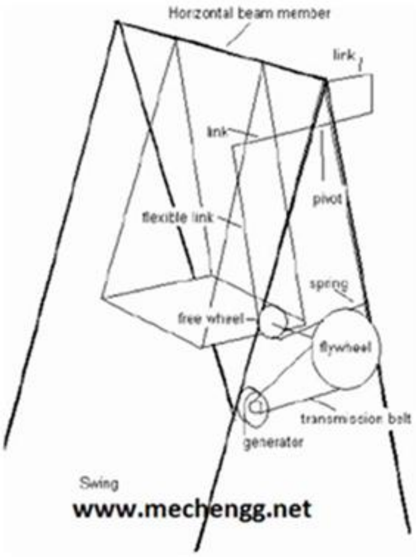
For this project two UCP210 pillow block bearings were selected as the most suitable bearing for supporting the horizontal shaft within the wooden frame. The pillow block bearing will be bolted directly onto the wooden frame, making installation simple. Since the shaft spans 1500mm between the two support points and carries a total load of 800N, the pillow block bearings provide the necessary load support at each end while allowing the shaft to rotate freely. The bearings can accommodate any slight misalignment that may occur during construction of the wooden frame. For this system two UCP210 pillow block bearings rated at 35kN will be used, one at each support point on the shaft.

3.3.10 Roller clutch bearings:

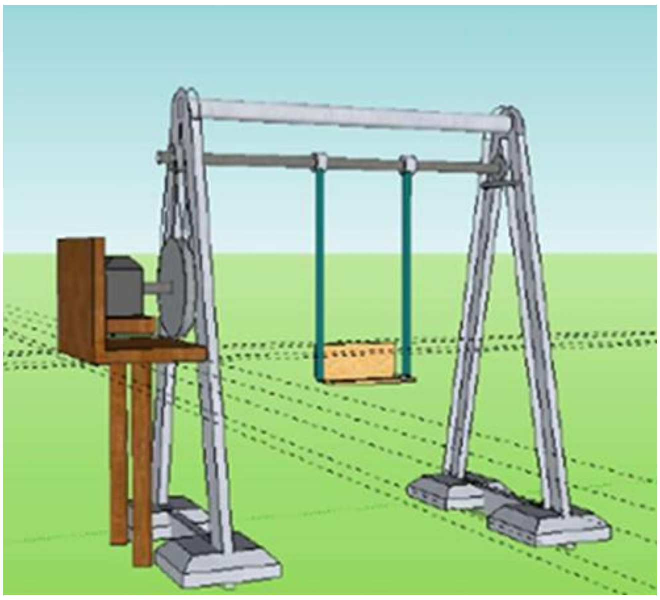
For this project two HFL5020 roller clutch bearings were selected as the most suitable bearing for connecting the swing to the horizontal shaft. The roller clutch bearing is a one way bearing that engages in one direction and freewheels in the opposite direction. When the swing moves in the forward direction the roller clutch bearing locks and transfers the motion to the shaft causing it to rotate, and when the swing returns in the opposite direction the bearing disengages and freewheels allowing the shaft to continue rotating in the same direction without being forced backwards. Two roller clutch bearings are used, one at each swing chain connection point on the shaft at positions B and C, ensuring that the load is evenly distributed across the shaft. For this system two HFL5020 roller clutch bearings with a 50mm bore diameter will be used.

3.4 Expected Design

Figure 4: Mechanical layout of the proposed swing generator



Swing Electricity Generation System Mechanical Project



3.5 Calculations:

Firstly, the output power should be determined in order to design the swing to meet the requirements. For demonstration purposes, the swing is required to have enough output power for small appliances. This system is designed to supply power sufficient to turn an LED on, phone charging and small appliances.

A standard LED consumes 5-10W [20], a phone charging via a standard USB draws approximately 10W [21] and small appliances such as a fan consume around 10-25W. Based on these values, a realistic load of 10W is the finalised output power of this system. This value is achievable by a human-powered swing generator.

3.5.1 Torque on the vertical shaft:

Taking note of the forces acting on the vertical shaft and the torque:

Downward force:

$$T = F_1 \times L$$
$$T = M \times g \times l$$
$$T = (M + m) \times g$$
$$= (100 + 1.5) \times 9.81$$
$$= 995.715\ N$$

Where:
M- maximum weight sitting on the swing seat
m- weight of the swing seat
L- distance of the swing seat to the vertical shaft

Torque force during movement:

$$T = F_1 \times L$$
$$T = M \times g \ \times l$$
$$\begin{aligned} T &= (M + m) \times g \\ &= (100 + 1.5) \times 9.81 \times \sin(45°) \times L \\ &= 80.5 \times 9.81 \times 0.7071 \times 1.1 \\ &= 705.170\ N \end{aligned}$$

Bending moment at B:

$$\begin{aligned} M_B &= R_A \times 500 \\ &= 750 \times 500 \\ &375\ 000 Nmm \end{aligned}$$

Equivalent Torque:

$$\begin{aligned} T_e &= \sqrt{M^2 + T^2} \\ &= \sqrt{200\ 000^2 + 705\ 170^2} \\ &= 798\ 980 Nmm \end{aligned}$$

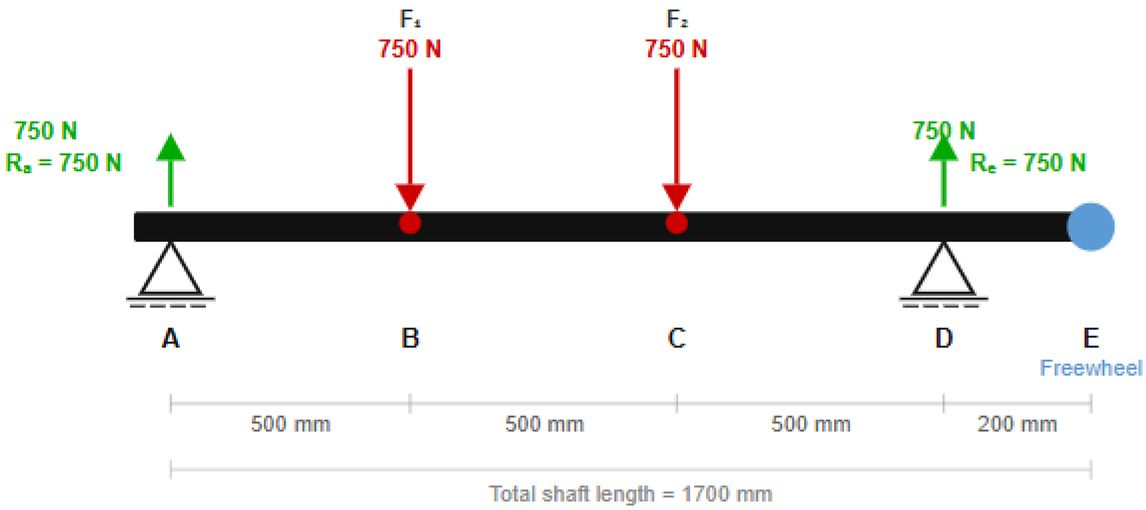


Figure 5: Diagram of forces acting on the support beam

Assume the weight of a person swinging (W)

$$W = M + m = 100 + 1.5 = 101.5\ kg$$
$$W = m \times g = (101.5 + 1.5) \ \times 9.81 = 799.515N = 800N$$

Therefore, load acting at B=1500/2= 750N
Load acting at C= 1500/2 = 750N

3.5.2 Shaft Diameter:

$$d^3 = \frac{32 \times \text{Te}}{\pi \times 89.3}$$
$$d^3 = \frac{25\,568\,000}{280\,530}$$
$$d = 45\ mm$$

Applying safety factor:

$$Safety\ factor = \frac{\sigma_{yield}}{\sigma_{allowable}}$$
$$= \frac{250}{89.3}$$
$$= 2.8$$

3.5.3 Generator rating:

To determine the rating of the generator:

$$f = \frac{1}{2\pi} \times \sqrt{\frac{g}{L}}$$
$$= \frac{1}{2\pi} \times \sqrt{\frac{9.81}{1.1}}$$
$$= 0.475\ Hz$$

Converting to RPM: $Swing\ RPM = 0.436Hz \times \frac{60sec}{min} = 28.5\ RPM$

Therefore, the swing completes 0.436 swings every second and in a full minute, it completes 28.5 full swings.

To find the required generator RPM, we need the generators voltage constant (Kv).

$$RPM = \frac{V_{output}}{K_v}$$

To be able to meet the requirement of powering a light, small appliance and charging cellphone, the output voltage $V_{output} = 12V$.

3.5.4 Sprocket sizing:

A driven sprocket needs at least 4-6 teeth to engage the chain without having faults or errors.

The required ratio can be determined by making use of the rated generated speed and the swing speed:

$$Ratio = \frac{N_{generator}}{N_{swing}}$$
$$= \frac{320}{28.5}$$
$$= Ratio = 13.3{:}\,1$$

Selecting freewheel and sprocket teeth:

The freewheel has 80 teeth and a driven sprocket of 6 teeth.

With this setup:

$$N_{generator} = 28.5 \times \left(\frac{80}{6}\right)$$
$$= 380\ RPM$$

This speed of 380 RPM will produce an output voltage of 14.25 V.
This exceeds the rated 12V, which is suitable for battery charging through the boost converter and charge controller.

3.5.5 Bearing selection:

Pillow Block bearing:
To find the dynamic load rating:

$$L10 = \left(\frac{C}{P}\right)^3 \times \left(\frac{10^6}{60n}\right)$$

Where:
L10- bearing life in hours
C- dynamic load rating
P-equivalent dynamic load
n- speed= 28.5 RPM

$$C = P\ \times \left(L10 \times \frac{60n}{10^6}\right)^{\frac{1}{3}}$$
$$= 400 \times 3.246$$
$$= 1.298kN$$
$$\approx 1.3kN$$

Roller Clutch bearing:

Torque capacity of each roller clutch bearing:

$$T_{each} = \frac{614.24}{2}$$
$$= 307.12Nm$$

3.5.6 Battery selection:

Table 3-8: Types of devices and their power ratings

Load device:	Power:
Phone charging	10W
LED light	10W
Small fan	10W

Assume worst case where one device is used for 5 hours:
$E = P \times t$
$= 10W \times 5$
$= 50Wh\ per\ day$

Capacity after Depth of Discharge (50%):

How much of the battery capacity is allowable to use before it needs recharging. Therefore, we will make use of the 50% rule to preserve battery life.

$$Ah = \frac{E}{V} = \frac{50}{12} = 4.17\ Ah$$

$$Ah_{rated} = \frac{4.17}{0.50}$$

$$= 8.33Ah$$

Now including 80% Efficiency:

$$Ah_{final} = \frac{8.33}{0.8} = 10Ah$$

Therefore, the 12V 10Ah Sealed Lead Acid battery is chosen.

3.6 List of finalised components:

- Swing frame (Consisting of 7 poles)
  - Leg poles: 4 Gum poles 2100mm, pole diameter of 75-100mm
  - Cross poles: 2 Gum poles 1700mm, pole diameter 75-100mm
  - Top support beam: 1 Gum pole 1500mm, pole diameter 75-100mm

- Horizontal Shaft (Rotating): Round bar 45mm * 1700mm
- Pillow block bearings: 2
- Roller clutch bearings: 40mm diameter, quantity=2
- Freewheel: 80 teeth, 25H
- Roller chain: 25H, pitch= 6.35mm
- Generator sprocket: 6 teeth, 25H
- DC Generator: DC Motor 12V 320RPM
- Full wave bridge rectifier: W10M Bridge Rectifier
- Boost converter: XL6009 Boost Converter Module
- Rechargeable battery: 12V 10Ah Sealed Lead Acid

3.7 Simulations:

The non-linear pendulum was simulated using Simulink. The Non-linear Damped Pendulum consists of an object that swings back and forth.

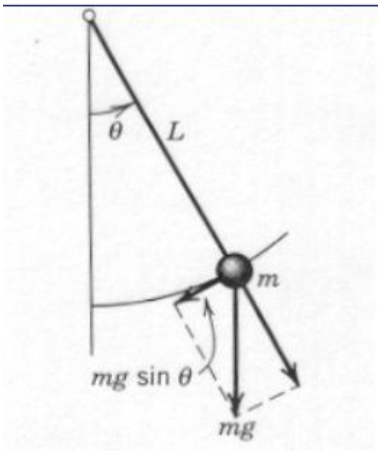


Figure 6: Diagram of a non-linear damped pendulum

A playground swing behaves similarly to a simple pendulum, where a mass **m** is attached to a rod of length **l**. As shown in figure 6, when the swing is displaces by angle θ from the vertical surface, the restoring force acting along it is:

$$F = -mgsin(\theta)$$

Applying Newtons second law of rotational motion:

$$mL^2\theta = -mgL\sin(\theta)$$

$$\theta = -\left(\frac{g}{L}\right) sin(\theta)$$

This is the nonlinear pendulum equation, where the sin(θ) shows that the angle between the vertical and the rod varies.

This equation was used in Simulink to simulate the motion of the swing movement.

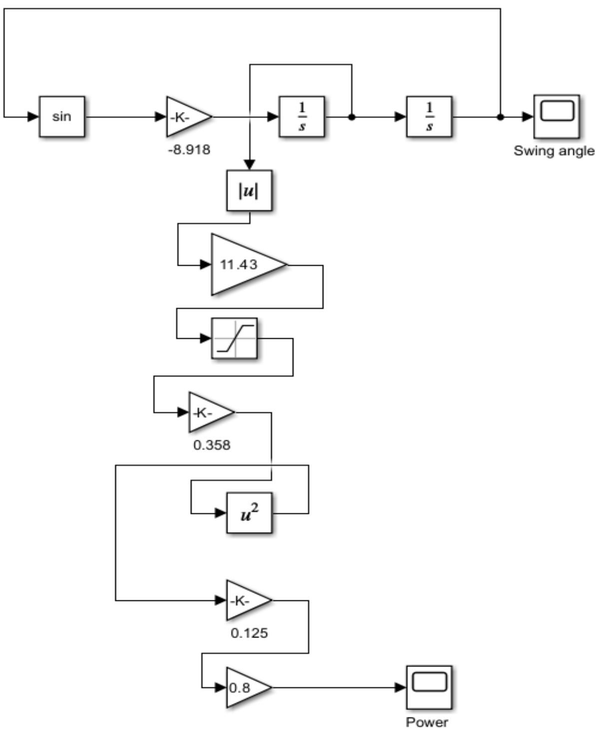


Figure 7: Simulation circuit of the swing mechanics

The Simulink model simulates the swing movement and the power output from the system. The model consists of two parts.
The first part models the swing as a nonlinear pendulum from the pendulum equation. The angular displacement θ is fed back through a sin block and multiplied by the gain -8.918, which represents -g/L. Two integrator blocks then produce the angular velocity $\dot{\theta}$ and angular displacement θ, which is fed back to close the loop. This feedback structure is what makes the model nonlinear.
The second part takes the angular velocity output and calculates the electrical power. The absolute value block handles the bidirectional swing motion due to the addition of a full wave rectifier, the ratio gain of 13.3 converts swing speed to generator speed, and the saturation block clamps the generator at its rated speed of 33.51 rad/s to prevent the motor from going past the motors limit. The back-EMF constant of 0.358, the squaring block and motor resistance of 0.125 produce the electrical power, and a final efficiency gain of 0.8 accounts for system losses. The output is displayed on the Power scope.

Figure 8: Plot of expected output power

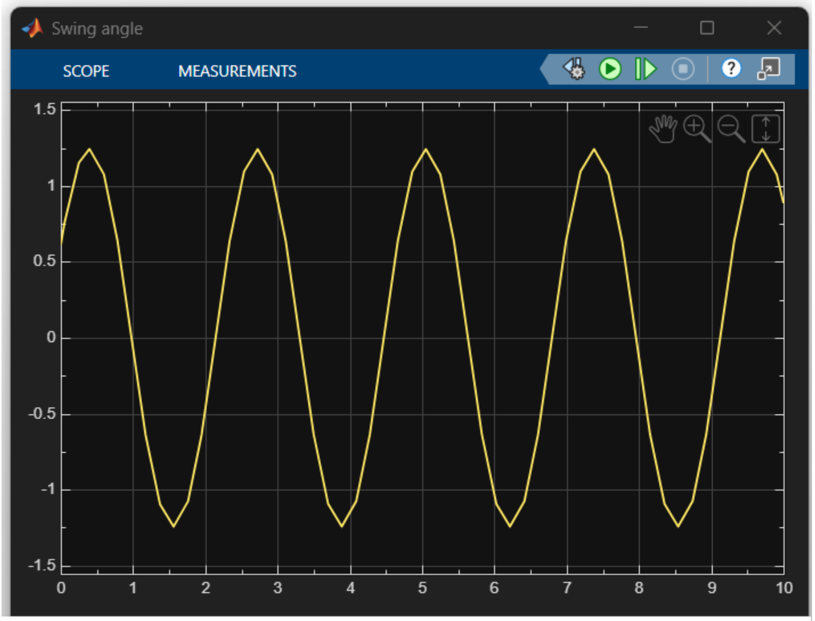


Figure 9: Plot of Swing angle

3.7.1 *Effect of Swing angle on Power output:*

Table 3-9: Swing angle effect on output power

Swing Angle:	Initial condition (radian):	Peak power (W):
20°	0.35	2.3W
45°	0.79	12W
60°	1.05	20W
90°	1.57	40W
120°	2.09	57W

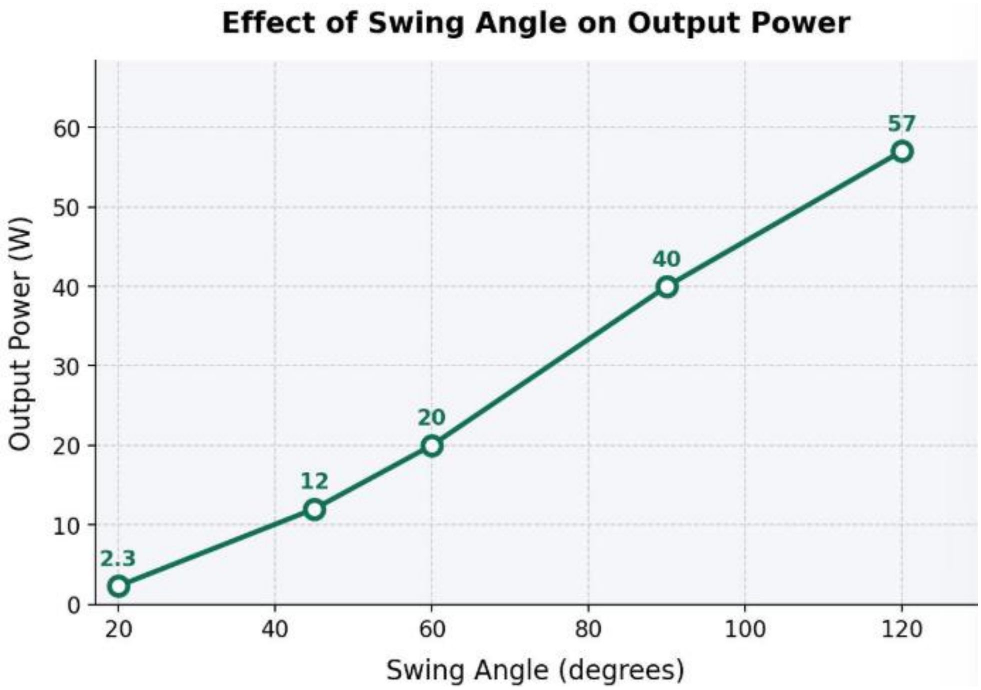


Figure 10: Plot of Output power VS swing angle

3.7.2 *Effect of Swing angle on Voltage:*

Table 3-10: Voltage due to varying swing angle

Swing angle:	Peak voltage:
20°	4.8

45°	11
60°	14
90°	20
120°	24

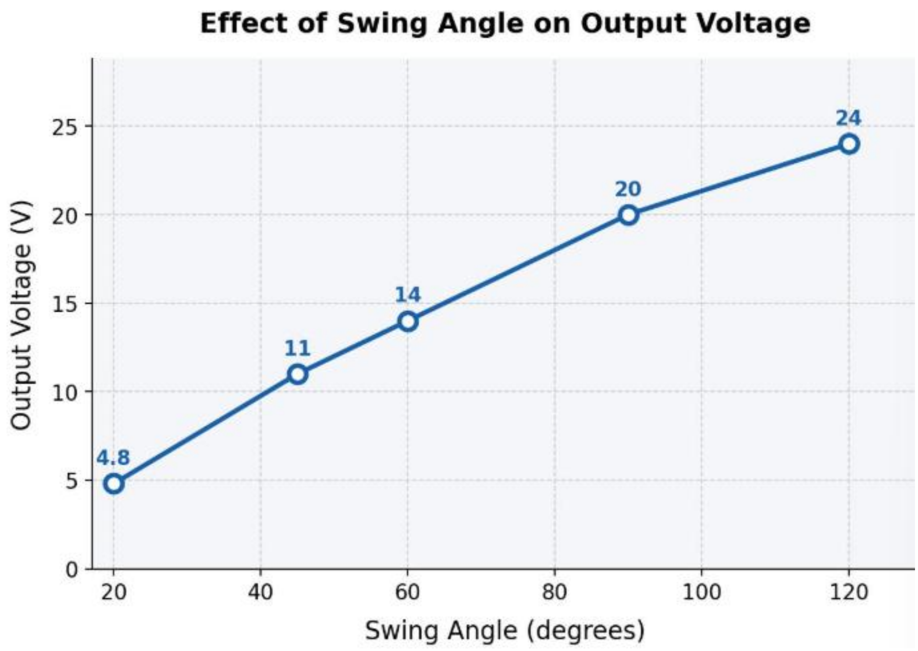


Figure 11: Plot of Output voltage VS Swing angle

3.7.3 Effect of Swing angle on Current:

Table 3-11: Effect of varying swing angle on peak current

Swing angle:	Peak power (W):	Peak voltage:	Peak current:
20°	2.3W	4.8	0.48
45°	12W	11	1.09
60°	20W	14	1.43
90°	40W	20	2.00
120°	57W	24	2.38

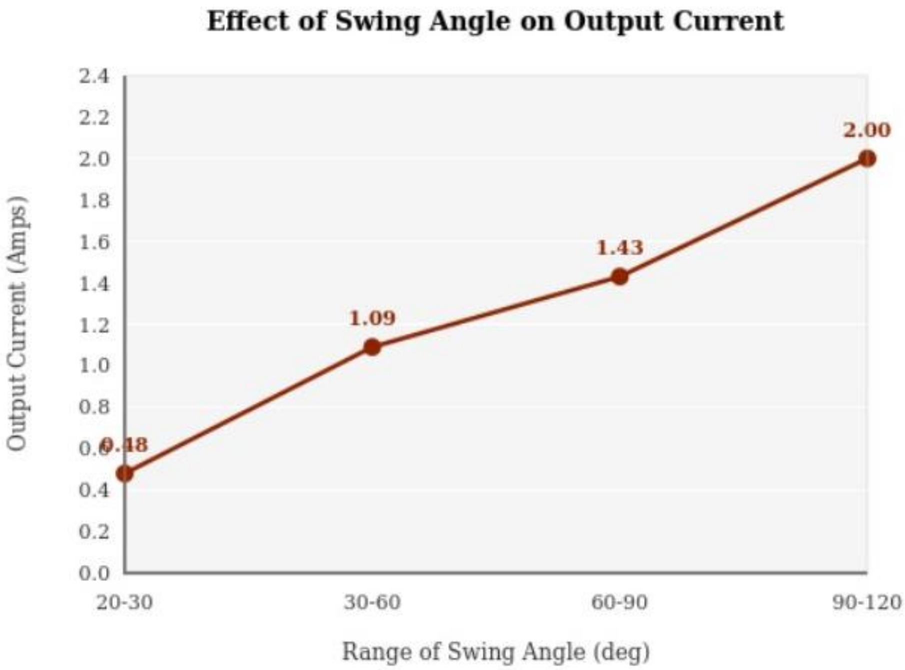


Figure 12: Plot of Output current VS Swing angle

From the simulation results, it can be seen that the swing angle has a direct effect on the output power, voltage and current. As the swing angle increases, the angular velocity at the bottom of each stroke increases, therefore driving the generator at a higher speed and producing greater power, voltage and current. At a low swing angle of 20°, the system produces only 2.3W, 4.8V and 0.48A, which is insufficient for any practical load. As the angle increases to 45°, the output improves significantly to 12W, 11V and 1.09A, meeting and exceeding the 10W design target. At 60° the system produces 20W, 14V and 1.43A, further confirming that the output continues to rise with increasing swing angle. These results demonstrate that the swing power generator is dependent on the effort and amplitude of the user's swing, and that an active swinging motion of at least 45° is necessary to generate sufficient electrical energy for practical low-power applications such as phone charging and LED lighting.

4. Expected Results

It is anticipated that the proposed swing-based electrical power generator will successfully convert oscillatory motion into usable electrical energy through the designed swing power generator system. The model is expected to show that the oscillatory motion of the swing can be effectively transformed into rotational motion which drives the generator.

The system is expected to produce a measurable electrical output sufficient to power low-energy devices such as LED lights or small electronic components. It is further expected that the output power will vary depending on factors such as the amplitude and frequency of the swing motion, as well as the efficiency of the mechanical conversion mechanism.

Additionally, simulation and experimental results are expected to demonstrate the overall performance of the system.

5. Plan of Development

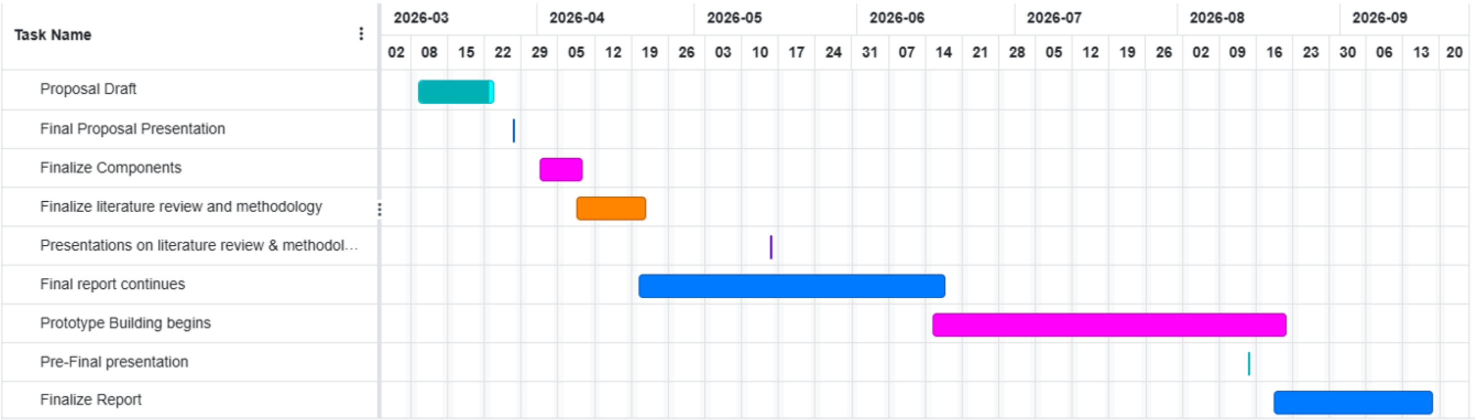


Figure 13: Gantt Chart of project

Summary of Gantt Chart:

- February, March: Proposal draft is written and proposal presentation.
- April, May: Finalising the literature review and methodology and finalise components as well as simulations. Presentation on literature review and methodology.
- April, May, June: Prototype building begins.
- September, October: Finalise report and pre-final presentation.
- October: Final project due.

6. Conclusion

In conclusion, this project proposes the development of a swing-based electrical power generator as an innovative approach to small-scale renewable energy harvesting applications. By converting human powered mechanical motion into electrical energy, the system aims to demonstrate a practical and sustainable energy solution. The proposed design includes kinetic energy conversion and electrical generation principles, supported by simulation and prototype testing. This project will aim to serve as an educational tool to help communities to understand renewable energy and how effective it is. It will also be a useful tool in powering low-power applications such as charging cell phones and powering lights.

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